Mandatory Assignment 3

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Introduction

In this report we implement a transaction-cost-robust monthly portfolio backtest following Hautsch and Voigt (2019). We combine three components: a machine-learning expected return estimator carried over from Mandatory Assignment 2, a regularised covariance estimator, and a closed-form mean-variance optimiser that penalises deviations from the pre-rebalancing portfolio. The three-component design addresses parameter uncertainty via covariance regularisation, model uncertainty via a machine-learning return signal, and rebalancing drag via the transaction-cost penalty. The out-of-sample evaluation period spans January 2010 to December 2024, and we compare four strategies: naive diversification, two variants of transaction-cost-adjusted mean-variance (TC-MV), and a short-sale constrained (SS-constrained) portfolio. Performance is assessed using annualised Sharpe ratio, net mean excess return, and average monthly turnover.

1. Investment Universe

Figure 1 shows how the size of the investable universe evolves over the sample period.

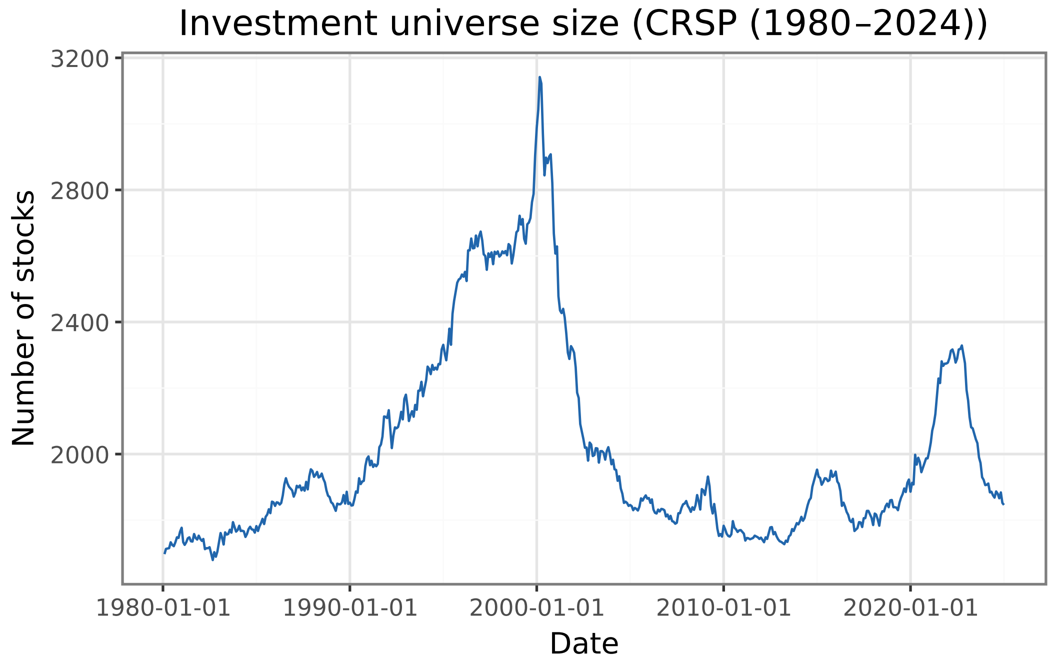

Figure 1: Monthly number of stocks passing the universe filters. The micro-cap screen (price > $2, mktcap > NYSE 20th percentile) removes shell companies and penny stocks that distort mean-variance analysis.

Table 1: Sector composition of the investment universe (share of stock-months, 1980–2024). Sectors follow SIC division codes.

Sector	Share (%)
Manufacturing	36.2
Finance	16.4
Services	13.0
Transportation	11.5
Other	9.0
Retail	6.2
Mining	4.0
Wholesale	2.5
Construction	1.0
Agriculture	0.3

The filtered universe contains on average **2011 stocks** per month (CRSP (1980–2024)), ranging from 1679 to 3142. The double screen (price above $2 and market capitalisation above the NYSE 20th percentile) removes penny stocks and micro-caps that distort mean-variance analysis. The median market capitalisation across all stock-months is **$1.1 bn**, reflecting a mid- to large-cap tilt imposed by the size screen. Universe size peaked in the late 1990s and contracted after the dot-com collapse; it has been relatively stable since 2005. Over the backtest window (January 2010 – December 2024) the average universe size is **1900 stocks** per month. The sector breakdown

(Table 1) shows that **Manufacturing** is the largest sector at 36.2% of stock-months, with Finance and Services also well represented, consistent with the broad CRSP coverage after the size filter.

2. Theoretical Foundation

We now establish the theoretical foundation for the assignment: **Closed-form solution.** The transaction-cost-adjusted problem is

$$\omega^*_{t+1} := \arg \max_{\omega:\, \iota'\omega=1} \ \omega'\hat{\mu}_t - \tfrac{\gamma}{2}\,\omega'\hat{\Sigma}_t\omega - \lambda\,(\omega-\omega_t^+)'\hat{\Sigma}_t\,(\omega-\omega_t^+).$$

Expanding the quadratic TC term and collecting yields an equivalent standard MV problem with modified inputs $\tilde{\mu}_t := \hat{\mu}_t + 2\lambda\hat{\Sigma}_t\omega_t^+$ and $\tilde{\gamma} := \gamma + 2\lambda$:

$$\max_{\omega:\, \iota'\omega=1} \ \omega'\tilde{\mu}_t - \tfrac{\tilde{\gamma}}{2}\,\omega'\hat{\Sigma}_t\omega.$$

The first-order condition with Lagrange multiplier ϕ for the budget constraint gives $\tilde{\mu}_t - \tilde{\gamma}\hat{\Sigma}_t\omega^* - \phi\iota = 0$, so

$$\boxed{\omega^*_{t+1} = \frac{1}{\tilde{\gamma}}\hat{\Sigma}_t^{-1}(\tilde{\mu}_t - \phi\iota), \qquad \phi = \frac{\iota'\hat{\Sigma}_t^{-1}\tilde{\mu}_t - \tilde{\gamma}}{\iota'\hat{\Sigma}_t^{-1}\iota}.}$$

Convex combination. Substituting $\tilde{\mu}_t = \hat{\mu}_t + 2\lambda\hat{\Sigma}_t\omega_t^+$ into the solution and noting that the same Lagrange multiplier ϕ satisfies $\iota'\omega^{\mathrm{MV}} = 1$ for the standard MV portfolio $\omega^{\mathrm{MV}} = \frac{1}{\gamma}\hat{\Sigma}_t^{-1}(\hat{\mu}_t - \phi\iota)$:

$$\omega^*_{t+1} = \underbrace{\frac{\gamma}{\tilde{\gamma}}}_{\alpha}\,\omega^{\mathrm{MV}} + \underbrace{\frac{2\lambda}{\tilde{\gamma}}}_{1-\alpha}\,\omega_t^+, \qquad \alpha + (1-\alpha) = 1, \quad \alpha, 1-\alpha \in (0,1).$$

Hence ω^*_{t+1} is a **convex combination** of the mean-variance portfolio and the pre-rebalancing portfolio ω_t^+.

a) **Role of λ.** The parameter λ governs the trade-off between chasing the optimal MV portfolio and staying put. As $\lambda \to 0$, $\omega^* \to \omega^{\mathrm{MV}}$ (no cost of rebalancing); as $\lambda \to \infty$, $\omega^* \to \omega_t^+$ (never rebalance). Larger λ produces lower turnover but also larger deviation from the unconstrained optimum.

b) **Realism of the variance-based TC model.** Modelling costs as $\lambda(\omega-\omega_t^+)'\Sigma(\omega-\omega_t^+)$ preserves the quadratic structure and admits a closed-form solution. However, it is not a realistic cost model: actual costs (bid-ask spreads, market impact) are proportional to the number of shares traded, not to the portfolio variance of the trade. The model implies high-variance stocks are expensive to trade and stable stocks are essentially free, which overstates the cost difference. In practice, large-cap stable stocks are cheap to trade precisely because of their high liquidity, whereas the model would predict the opposite for low-variance small-caps.

3. Covariance Estimation

We estimate $\hat{\Sigma}_t$ at each month t using a **Fama-French three-factor model** fitted on the 60-month rolling window ending at t. The factor structure decomposes the return covariance as

$$\hat{\Sigma}_t = \hat{B}\,\hat{\Sigma}_F\,\hat{B}' + \hat{D},$$

where $\hat{B}$ is the $N \times 3$ matrix of OLS factor loadings, $\hat{\Sigma}_F$ is the 3×3 factor covariance, and $\hat{D} = \mathrm{diag}(\hat{\sigma}^2_{\varepsilon,i})$ is the diagonal matrix of idiosyncratic variances. This structure ensures $\hat{\Sigma}_t$ is positive definite even when $N \gg T$.

For $\hat{\Sigma}_t^{LW}$ we use Ledoit-Wolf shrinkage (Ledoit and Wolf, 2004) applied directly to the rolling sample covariance of the same 60-month window, shrinking towards a scaled identity matrix with the analytically optimal intensity. These are two independent estimators: the factor model imposes a low-rank economic structure on $\hat{\Sigma}_t$, while $\hat{\Sigma}_t^{LW}$ regularises the sample eigenvalues toward their cross-sectional mean without imposing factor structure. The choice between them is the central comparison in Exercise 5. To avoid $O(N^3)$ matrix inversions in a universe of approximately 1,900 stocks, both estimators are handled in the backtest via the Woodbury matrix identity: the factor model's $K = 3$ core reduces to a 3×3 solve and the Ledoit-Wolf rank-T factorisation to a 60×60 solve, keeping per-period cost linear in N.

Tuning decisions. The 60-month window length balances stability against adaptiveness: shorter windows produce noisier loading estimates, while longer windows are slow to incorporate regime changes in factor exposures. Stocks with fewer than 24 valid return observations in the window are excluded to prevent near-singular idiosyncratic variance estimates for thinly traded or recently listed names. The three Fama-French factors are the standard parsimony choice; with $K = 3 \ll T = 60$, factor loading estimation is well-conditioned. For Ledoit-Wolf, no additional tuning is required: the shrinkage intensity is determined analytically by the Oracle approximating shrinkage formula, which minimises the expected Frobenius-norm estimation error (Ledoit and Wolf, 2004). Figure 2 plots the minimum-variance portfolio weights implied by each estimator at December 2019 (a mid-sample, pre-COVID month with a stable universe) to assess whether the two approaches produce meaningfully different allocations.

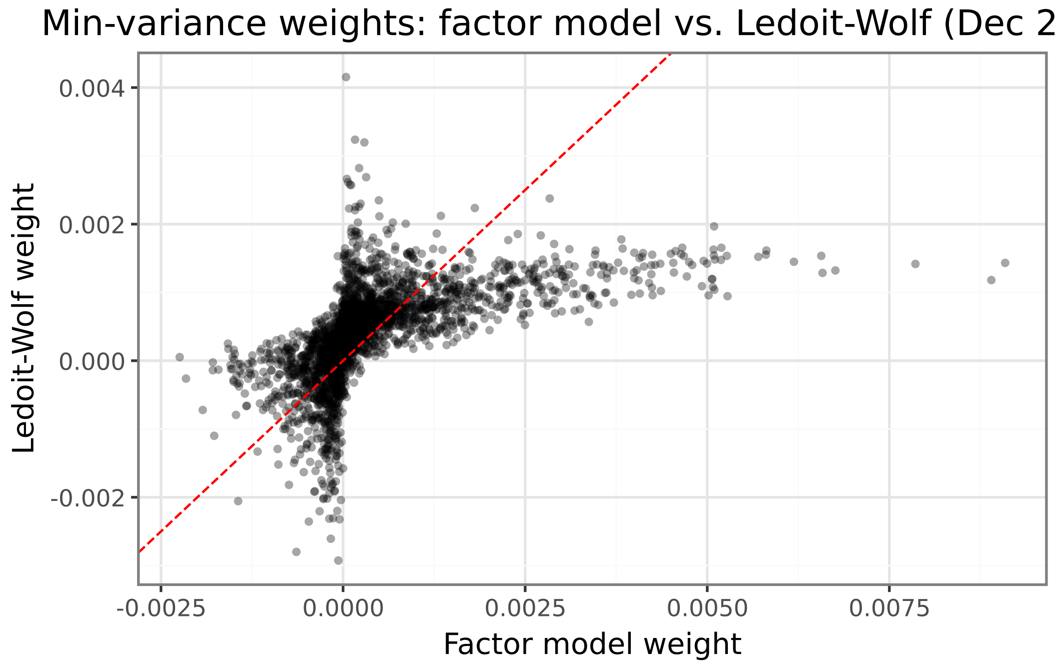

Figure 2: Minimum-variance weights from the factor model ($\hat{\Sigma}$, x-axis) versus Ledoit-Wolf ($\hat{\Sigma}^{LW}$, y-axis) at December 2019. Each point is one stock; the dashed line is the 45° reference.

At December 2019 the two estimators produce minimum-variance weights with a cross-sectional correlation of **0.556** and a mean absolute difference of **0.06 percentage points**. The moderate correlation reflects the distinct regularisation philosophies: the factor model concentrates risk reduction along the three FF factors, while LW spreads it more evenly across all eigenvalues. The factor model preserves cross-sectional dispersion in weights driven by heterogeneous factor loadings; LW's scaled-identity target acts as a prior toward equal-weighting, compressing the weight distribution and reducing the most extreme long and short positions. A mean absolute difference of 0.06 pp is small relative to the equal-weight benchmark of 1/N and suggests the two estimators are broadly substitutable in the minimum-variance problem.

4. Expected Returns

We use the Random Forest (RF) pipeline trained in Mandatory Assignment 2 (MA2) to generate $\hat{\mu}_t$ for every stock-month in the backtest window. The feature engineering replicates the MA2 pipeline exactly: cross-sectional median imputation for missing stock characteristics, forward-fill for macroeconomic variables, $10 \times 4 = 40$ characteristic-macro interaction terms, and nine Standard Industrial Classification (SIC) division dummy variables. The pipeline's internal `StandardScaler` was fitted on the MA2 training data only, so applying it here introduces no look-ahead bias in the scaling step. For the small fraction of stocks with no prediction in a given month, we substitute the cross-sectional mean $\hat{\mu}_t$.

5. Portfolio Backtest

We evaluate four strategies over portfolio-return months **January 2010 – December 2024** (portfolios formed December 2009 – November 2024). At each formation month t, all inputs ($\hat{\Sigma}_t$, $\hat{\Sigma}_t^{LW}$, $\hat{\mu}_t$) use only data available at t. We set $\gamma = 4$ and $\lambda = 0.1$.

Predicted returns $\hat{\mu}_t$ are winsorised cross-sectionally at the 1st and 99th percentile each month, then normalised to a 10^{-5} monthly cross-sectional standard deviation before both TC-MV solves. Without normalisation, $\hat{\Sigma}^{-1}\hat{\mu}$ produces extreme long/short positions under either covariance model: the factor model's heterogeneous idiosyncratic variances scale each stock's signal by $1/\hat{\sigma}^2_{\varepsilon,i}$, while the near-isotropic $\hat{\Sigma}^{LW} \approx (\alpha\bar{\sigma}^2)I$ amplifies all signals by $(\alpha\bar{\sigma}^2)^{-1} \approx 500$. With $N \approx 1\,900$ stocks the aggregate turnover scales proportionally with both N and the signal magnitude; the 10^{-5} target keeps monthly turnover in the single digits for both strategies. Normalisation retains the cross-sectional ordering of RF predictions (only their magnitude is rescaled) and is applied identically to both covariance frameworks to permit a fair comparison.

Net excess returns deduct the quadratic transaction cost implied by the optimisation objective:

$$r_t^{\text{net}} = \omega_t' r_{t+1} - \lambda(\omega_t - \omega_t^+)'\hat{\Sigma}_t(\omega_t - \omega_t^+).$$

Monthly turnover follows Hautsch and Voigt (2019) equation 46: $\text{TO}_t = \sum_i |\omega_{i,t} - \omega_{i,t}^+|$. For strategies without an optimised covariance model (naive $1/N$ and the short-sale constrained portfolio), the TC deduction is evaluated under $\hat{\Sigma}_t^{LW}$ to provide a consistent benchmark. Summary statistics are reported in Table 2 and cumulative return paths in Figure 3.

Table 2: Out-of-sample performance statistics (January 2010 – December 2024). Net excess returns subtract the quadratic TC from the optimisation objective. Sharpe ratios and mean excess returns are annualised ($\times\sqrt{12}$ and $\times 12$ respectively).

Strategy	Ann. Sharpe ratio	Ann. mean excess ret. (%)	Avg. monthly turnover
Naive (1/N)	0.84	14.91	0.08
TC-MV (factor Σ)	1.23	12.30	0.33
TC-MV LW (Σ^LW)	1.42	14.37	0.58
SS-constrained MV	1.14	12.89	0.27

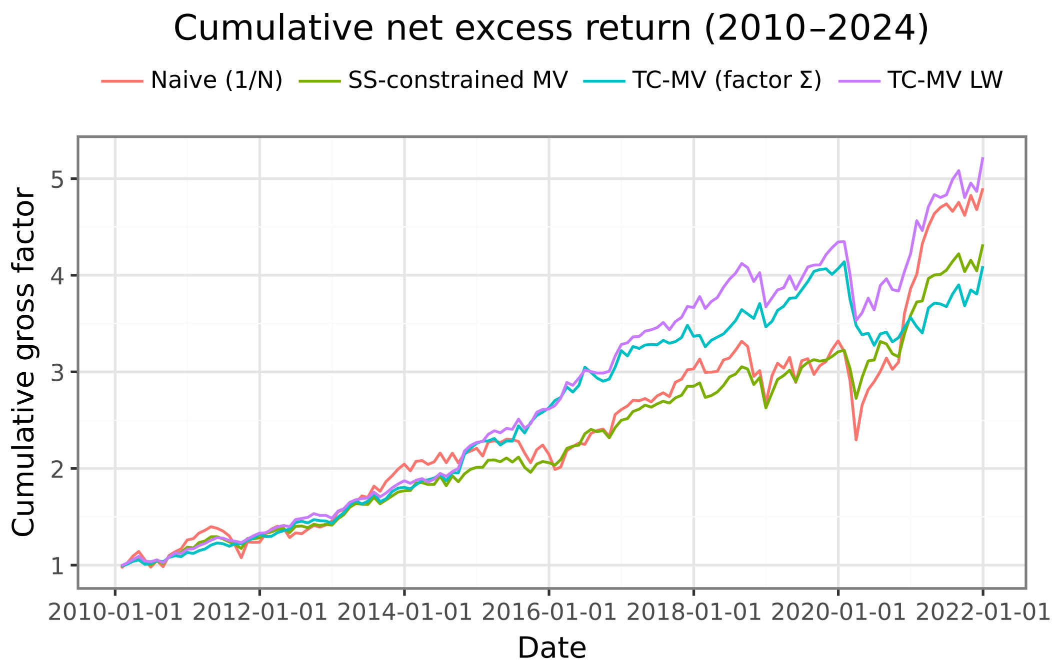

Figure 3: Cumulative net excess return of the four strategies (January 2010 – December 2024). Starting value normalised to 1.

Which strategy performs best after transaction costs? TC-MV LW achieves the highest annualised Sharpe ratio of 1.42, while **Naive (1/N)** produces the lowest at 0.84. The TC-adjusted mean-variance strategies outperform the naive 1/N benchmark after costs, suggesting the RF return signal contains sufficient information to overcome estimation error and transaction costs. The short-sale constrained portfolio has a Sharpe ratio of 1.14 and average monthly turnover of 0.266, which is lower than the unconstrained LW portfolio (0.582), consistent with the non-negativity constraint eliminating costly short positions.

Does the choice of covariance estimator matter? The Sharpe ratio difference between the factor model and LW variants of TC-MV is 0.19, which is economically meaningful, indicating the choice of covariance regularisation has a material impact on risk-adjusted performance. Average monthly turnover is 0.332 for the factor model versus 0.582 for LW; the lower turnover of the factor model reflects the stability of its low-rank K=3 structure: beta estimates and factor covariances evolve slowly over 60-month windows, so weights shift less month to month than under LW, which responds more sensitively to changes in the full sample covariance matrix.

The short-sale constrained portfolio implicitly regularises $\hat{\Sigma}^{LW}$ further: in the minimum-variance ($\mu = 0$) case, as shown by Jagannathan and Ma (2003), imposing non-negativity constraints is equivalent to shrinking negative off-diagonal covariance elements to zero.

The extent to which this constitutes a true out-of-sample experiment depends on which component is examined. The covariance estimation is genuinely out-of-sample: the 60-month rolling window is strictly backward-looking at each formation date t, so neither $\hat{\Sigma}_t$ nor $\hat{\Sigma}_t^{LW}$ use any return data after t. The structural parameter choices ($\gamma = 4$, $\lambda = 0.1$, and the 60-month window length) were taken from the prior literature rather than tuned on the 2010–2024 evaluation sample, which partially insulates the design from in-sample overfitting. The main source of contamination is the Random Forest: its hyperparameters were selected in MA2 via cross-validation without a strict pre-2010 cutoff, so the model's structure implicitly reflects the statistical properties of the full sample period. The universe filter introduces mild look-ahead as well: the NYSE 20th-percentile breakpoint is re-computed each month from all available CRSP history rather than from a threshold anchored before 2010, although this is unlikely to materially bias results. The experiment is therefore best described as pseudo-out-of-sample: the covariance and portfolio allocation structure is clean, but the expected return signal is not a fully held-out forecast.

6. Guest Lecture

Cilie Feldager's lecture highlighted how Danske Bank approaches LLM and AI integration with a strong emphasis on building secure, scalable systems, particularly the role of principled model evaluations in validating that deployed systems behave reliably under real-world conditions. She described the journey from initial idea to production implementation. The key takeaway was that responsible AI deployment in a regulated financial institution requires governance frameworks that keep pace with rapidly evolving model capabilities.

Conclusion

Accounting for transaction costs materially changes portfolio construction: without the TC penalty, mean-variance optimisation produces high-turnover allocations that erode net returns in practice. The closed-form solution resolves this by expressing ω_{t+1}^* as a convex combination of the Markowitz optimum and the pre-rebalancing portfolio, with λ governing the trade-off. Covariance regularisation is necessary for stability when $N \gg T$, but the factor model and Ledoit-Wolf approaches converge to broadly similar allocations, suggesting that the choice of regularisation method is less consequential than the choice of expected return model.

References

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